

## Highlights

### **Does AI rate through the cycle? Procyclicality in credit assessment**

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- First controlled test of procyclicality in LLM credit ratings.
- All seven models downgrade identical firms by 0.18 to 0.78 notches per severity step.
- Frontier models do not differ significantly from a lower cost Flash model.
- The effect is asymmetric: downside sensitivity exceeds upside in every model.
- A “through-the-cycle” prompt anchors ratings rather than de-cyclifying them.

# Does AI rate through the cycle? Procyclicality in credit assessment

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## ARTICLE INFO

### *Keywords:*

large language models  
credit ratings  
procyclicality  
through-the-cycle  
prompt framing  
financial regulation

## ABSTRACT

Large language models (LLMs) are increasingly proposed as credit-rating and default-assessment tools, yet whether they rate “through the cycle” or amplify it has been asserted rather than measured. We hold synthetic firm fundamentals byte-identical across an ordered five-point macroeconomic-severity axis (plus a matched placebo axis) and elicit letter ratings and one-year default probabilities from seven models spanning two families from the lowest cost tier to the capability frontier, so that any systematic rating movement along the macro axis is procyclicality by construction, identified without the unobserved-fundamentals confound that limits the classical Amato–Furfine test. Every model is significantly procyclical, from 0.182 to 0.775 rating notches of downgrade per severity step, and 18 of 21 model pairs differ significantly. Capability lowers the level but not the character: neither frontier model differs significantly from a lower cost Flash-tier model, and downside sensitivity exceeds upside in every model. An explicit “through-the-cycle” instruction does not restore through-the-cycle behaviour. It anchors the rating while suppressing the point-in-time default probability that should keep moving, and it fails hardest at the frontier, where the rating flattens to zero or below and so does default-probability retention. A vaguer “be stable” instruction preserves the distinction far better, pointing to cycle-specific wording, not stabilisation as such. The pattern survives on real SEC-filer fundamentals, and the capital consequence keeps its sign across every mapping we vary. Among the tested models, greater capability does not eliminate this channel: it produces the most stable-looking rating while carrying the least risk signal.

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## 1. Introduction

Credit ratings are important for debt pricing, investment decisions, and regulatory capital requirements. A perennial concern is that ratings could be procyclical, with companies downgraded during recessions and upgraded during expansions even when the underlying credit fundamentals have not materially changed. Such rating changes can exacerbate the credit cycle by raising borrowing costs and, through internal-ratings-based (IRB) capital rules, capital requirements when economic conditions are already weak (Kashyap and Stein, 2004; Gordy and Howells, 2006). Rating agencies therefore attempt to capture creditworthiness “through the cycle” (TTC), placing more importance on a borrower’s ability to withstand temporary macroeconomic fluctuations than on mechanically reacting to current conditions (Amato and Furfine, 2004; Löffler, 2004). However, whether these ratings actually serve this purpose remains debated (Löffler, 2013).

There is growing interest in using large language models (LLMs) for financial decision-making, including credit decisioning and risk assessment, and supervisors have flagged that generative models could re-introduce procyclicality (Aldasoro, Gambacorta, Korinek, Shreeti and Stein, 2024). Institutional use is already emerging: Moody’s offers AI-powered systems for company credit assessment, underwriting, and credit-memo generation, with documented adoption by banks such as Taishin International Bank, while S&P Global has launched an agentic-AI Credit Memo Builder for loan committees, underwriters, and credit analysts (Moody’s, 2026a,b; S&P Global, 2026). These systems retain analyst oversight rather than replacing final credit judgment, but their deployment makes LLM behaviour directly relevant to these workflows. LLMs behave as context-sensitive economic agents (Horton, 2023) and can exhibit framing sensitivity (Binz and Schulz, 2023), echoing classical framing effects (Tversky and Kahneman, 1981), and a shared foundation model could homogenise decisions across users (Bommasani, Creel, Kumar, Jurafsky and Liang, 2022). Yet the size of this effect on credit ratings has not been directly measured. In observational studies of human agencies, fundamentals and macroeconomic conditions move together, so the classical test of Amato and Furfine (2004) must proxy for unobserved fundamentals.

This paper uses a controlled experimental setting to test the procyclicality of LLM-based credit rating. We keep the firm fundamentals unchanged and only change the macroeconomic environment, so any systematic change in the assigned rating can be attributed to the macroeconomic context and not to changes in firm credit quality. The benchmark is not that any movement is bad: a genuine through-the-cycle rater should hold the *rating* steady while its point-in-time default probability still tracks conditions, and we test that joint condition directly. We make three contributions. First, all seven models, including the frontier tier of both families, downgrade identical firms as conditions deteriorate. The level falls with capability, but neither frontier model differs significantly from a Flash-tier model. Second, the response is strongly asymmetric, with downside sensitivity exceeding upside sensitivity in every model. Third, prompt design does not fix it: an explicit through-the-cycle instruction flattens the rating while suppressing the one-year default probability's responsiveness, so the model anchors both outputs rather than separating point-in-time risk from a through-the-cycle rating, echoing classical anchoring effects (Tversky and Kahneman, 1974), and it does this most completely at the frontier.

## 2. Related literature

Our study relates to three strands. The first assesses through-the-cycle behaviour and rating procyclicality: whether ratings respond to the business cycle once fundamentals are controlled for, and whether agencies in fact achieve through-the-cycle stability (Amato and Furfine, 2004; Löffler, 2004, 2013). Rating levels also move market pricing directly (Cantor and Packer, 1996). The prudential stakes come from the interaction of ratings with Basel capital rules (Kashyap and Stein, 2004; Gordy and Howells, 2006).

The second strand is the rapid adoption of *LLMs in finance* (Ali, Zafar and Aysan, 2025), spanning return prediction, financial-statement analysis, financial advice and simulated economic agents (Lopez-Lira and Tang, 2023; Kim, Muhn and Nikolaev, 2024; Fieberg, Hornuf, Meiler and Streich, 2025; Horton, 2023) and, most relevant here, reliability evidence for banking supervision (García-Llorente and Olmeda, 2026), with related work measuring policy uncertainty and AI-firm financing (Ito, Sato and Ota, 2025; Giaretta and Zen, 2026). Most closely, Hens and Nordlie (2025) compare the risk profiles LLMs produce for investors with those of experienced bankers, finding relatively small differences in the levels of risk assigned, although the models give weaker explanations. We examine a different dimension of reliability: whether an LLM alters its credit assessment in response to rating-irrelevant changes in the macroeconomic context surrounding otherwise identical firm information.

Anchoring, framing and elicitation-dependent calibration are documented in language-model decisions (Binz and Schulz, 2023; Kadavath, Conerly, Askell et al., 2022), echoing the classical human framing literature (Tversky and Kahneman, 1974, 1981), and shared foundation models can homogenise decisions across users (Bommasani et al., 2022). In our setting these behavioural properties connect to a specific financial mechanism: if the macroeconomic context affects credit ratings while firm fundamentals are held fixed, contextual sensitivity translates directly into rating procyclicality.

## 3. Design and identification

The unit of analysis is a rating cell = (model, prompt framing, seed, firm, macro state). We construct 80 synthetic firms spanning five credit-quality tiers and eight sectors, and characterise each firm using the same set of credit fundamentals: revenue growth, EBITDA margin, leverage, interest coverage, free cash flow, liquidity, and Altman-style financial ratios (Altman, 1968). The information about each firm presented to the models is the same in all experimental conditions. Each firm's fundamentals are serialised once and reused byte-identically under every macro state. Following this, we merely alter the model's exposure to the macroeconomic environment by switching it between five ordered states: boom, expansion, neutral, slowdown, and severe recession (severity  $-2 \dots +2$ ). A common set of macroeconomic indicators is used to represent each state, including GDP growth, unemployment, credit spreads, default rates, and lending standards. No calendar dates are given, reducing the chance that a model associates a situation with a particular historical event.

To strengthen identification, we add two controls. We first generate a matched placebo condition carrying credit-irrelevant information about regional weather and traffic. The placebo matches the macro scenarios

in format and numerical density but is credit-irrelevant, so it separates macroeconomic sensitivity from a general tendency to respond to changing context. The net (macro minus placebo) coefficient is the demand-effect-adjusted cyclicity. Second, a firm and framing keep the same seed across all macro states, so sampling variation is not systematically related to the macro state. Each assessment produces a letter credit rating (mapped to an S&P notch on a 21-point scale) and a one-year probability of default (PD).

We consider four prompt framings: a terse baseline, a point-in-time instruction, an explicit through-the-cycle instruction, and a generic “be stable” instruction that makes no reference to the credit cycle. The experiment uses seven models spanning the full capability range of two families: four from OpenAI (**gpt-4o-mini**, **gpt-4.1-mini**, **gpt-4o**, **gpt-6-astra**) and three Google (**gemini-flash-lite-latest**, **gemini-flash-latest**, **gemini-pro-latest**). Three OpenAI arms call dated snapshots. The other four use undated aliases that cannot be version-resolved (Appendix A). Each cell is run at temperature zero where supported, which minimises decoding variability and isolates the macro axis as the treatment contrast, with replicate seeds: the primary condition carries five draws per cell for the four models whose additional-seed captures completed (**gpt-4o-mini**, **gpt-4.1-mini**, **gpt-4o**, **gemini-flash-latest**) and two for the remaining three (**gemini-flash-lite-latest**, **gemini-pro-latest**, **gpt-6-astra**), whose extra-seed captures exceeded the pre-registered 2% error-rate gate, which refuses a whole capture rather than individual observations. Two frontier-model properties constrain what can be held fixed: **gpt-6-astra** refuses temperature zero and runs at its default, and both frontier models reason internally by construction. Because decoding randomness is unrelated to macro severity, it should widen intervals rather than generate a systematic slope. **gemini-pro** does accept temperature zero, so the frontier result is reproduced at the design’s own decoding setting. The dependent variable is the change in rating notch as severity rises, positive meaning more procyclical. Intervals are 2,000-draw firm-clustered, tier-stratified bootstraps. Tables 1 and 4 and Fig. 2 report the secondary estimands: swing, half-slopes, the Amato–Furfine ordered probit and the log-probability expected-notch slope where the provider exposes it.

## 4. Results

### 4.1. LLMs are procyclical on identical fundamentals

On byte-identical fundamentals, all seven models downgrade the same firms as the described macroeconomy worsens, and no model’s interval contains zero (Table 1 and Fig. 1a). At the pooled terse coefficient (Table 2) a BBB firm moves toward BBB−/BB+ with nothing real changing.

Model choice is a first-order determinant: under a paired firm-clustered bootstrap 18 of the 21 model pairs differ significantly, so the fourfold spread in Table 1 is tested rather than inferred from overlapping intervals. Its structure matters more than its width. Within each family the coefficient declines monotonically in point estimate with capability, yet capability does not order it *across* families: the two tested frontier models converge near 0.19 from very different starting points rather than approaching zero. Before correction the only three pairs that do *not* differ are Gemini Flash versus Gemini Pro, Flash versus GPT-6 Astra, and Pro versus Astra: neither frontier model differs significantly from a Flash-tier model. Capability buys a lower level, not a different character, and Section 4.3 shows the character worsens.

The result survives the checks in Table 4: the credit-irrelevant placebo moves ratings only 0.0425 notches per step across the five original models, leaving a net effect of 0.387 (95% CI [0.363, 0.412]) that still excludes zero, and the coefficient holds under served-model fixed effects, a log-probability estimator and the Amato–Furfine ordered-probit estimator.

Magnitude is therefore a model-selection variable: the swings in Table 1 span over fourfold, frontier models under one notch and GPT-4o-mini above three. The claims we press are the structural ones, which do *not* attenuate with capability (§4.3).

### 4.2. The procyclicality is asymmetric

Split at neutral, downgrade sensitivity is about 3.2 times upgrade sensitivity pooled, and every model’s downside-to-upside interval excludes 1 (Table 1 and Fig. 2). That is a direction, not an ordering: the ratio’s denominator is small, so the per-model ranking is not resolvable. The direction is what matters, because downgrades in downturns raise borrowing costs and capital requirements exactly when conditions are already

strained. Because capital requirements are convex in the rating and bind hardest in stress, an asymmetric downgrade profile is more procyclical than a symmetric one of equal average slope.

### 4.3. “Through-the-cycle” instructions anchor rather than de-cyclify

Prompt framing is a strong determinant of rating sensitivity: the pooled slope falls monotonically across the four framings, roughly threefold from point-in-time to explicit through-the-cycle (Table 2). At first sight that suggests through-the-cycle prompting reduces procyclicality. It does not. Genuine through-the-cycle behaviour is a *joint* condition: the rating slope should fall toward zero *and* the point-in-time default probability should keep moving with the cycle. Neither is required to be exact, so we operationalise both as tolerance bands tested against their intervals: a rating slope within  $\pm 0.05$  notches per step of zero, and PD retention of at least 0.75 (Appendix A). We measure the default probability on the log-odds scale, which reduces dependence on PD *level* differences between models. That is not an external calibration test, and the calibration of verbalised LLM probabilities is task- and elicitation-dependent (Kadavath et al., 2022). We therefore do not rest the anchoring result on the elicited PDs alone. It is supported jointly by three measures: the rating-token log-probability slope, which never reads a stated PD and falls comparably under the instruction wherever the provider exposes it (Table 4), PD retention, which uses the elicited one-year PDs on the log-odds scale, and the “be stable” versus through-the-cycle control, which holds the elicitation fixed and varies only the instruction.

No model satisfies both conditions, and they fail in opposite directions (Table 1). The five smaller models fail the rating condition while retaining part of the PD slope. The frontier models invert that: Gemini Pro alone has a residual interval inside the band and retains none of the PD slope, while GPT-6 Astra overshoots it, not merely ceasing to track the cycle but reversing. Neither is through-the-cycle behaviour. The instruction stabilises the rating and suppresses the default probability that should remain sensitive: it anchors the wrong object, and most completely where capability is highest. The vaguer “be stable” instruction, which is not semantically about the cycle, leaves the rating slope higher while preserving far more default-probability movement, consistent with the additional suppression arising from cycle-specific wording rather than from stabilisation as such.

### 4.4. Capital consequences

What matters is not that the rating moves but what the movement costs. We translate the rating path into capital by anchoring through the *rating*, which is what enters a credit pipeline rather than the disclaimed verbalised PD: we map each notch to a stylised one-year default probability loosely calibrated to published one-year corporate default rates (S&P Global Ratings, 2025), and apply the Basel internal-ratings-based (IRB) corporate risk-weight formula (Basel Committee on Banking Supervision, 2020a) as a proxy for the economic logic (parameters in Appendix A). Reporting the *change* across the severity axis, an identical firm’s required capital rises, across the five original models, by a pooled 1.81 percentage points of exposure from boom to severe recession (95% CI [1.67, 1.94]), and by nearly twice that for the most sensitive model. Because the risk weight is convex in the rating, the §4.2 asymmetry compounds: the downside increase is about five times the upside decrease, a wider gap than the notch asymmetry alone. A 259-cell sensitivity grid leaves the sign positive in every cell (Appendix A), so the direction is robust while the magnitude is assumption-dependent.

## 5. Conclusion

With fundamentals fixed by construction, all seven models, including the frontier tier of both families, downgrade identical firms as conditions are described as worsening, and do so asymmetrically, punishing simulated downturns far more than they reward simulated expansions. The level of the effect falls with capability. Its structure does not. Neither frontier model differs significantly from a Flash-tier model, so the choice of model remains a prudential decision, but a more capable model is not an escape from the behaviour.

Prompt design mitigates it only partially, and at the frontier not in the way that matters. An explicit through-the-cycle instruction flattens the rating to zero or below for both frontier models, while driving the one-year default probability’s cyclical sensitivity to zero or below. The models stabilise both outputs instead of separating a through-the-cycle rating from point-in-time risk, and they do so most completely

**Table 1**

Per-model results. **Panel A:** procyclicality on byte-identical fundamentals.  $\beta$  = notches of downgrade per +1 macro-severity step under the terse framing, with firm-clustered bootstrap 95% CI; **Swing** = measured boom→severe-recession notches (not  $4\beta$ : the endpoint ratio runs 3.96–4.39, so the gap carries curvature); **D:U** = downside over upside half-slope. Fig. 2 plots the two half-slopes with their intervals. The ratio has its own clustered intervals, with lower bounds from 1.23 to 3.06 and upper bounds reaching 8.50. All exclude one, and they overlap across models, so they do not rank them. **Resid.** and **PD-ret.** =  $\beta$  and the retained fraction of the log-odds PD slope under the through-the-cycle instruction;  $\beta$  **real** = the same coefficient on the real-fundamentals battery and  $\Delta\beta$  its drift on re-measurement 2.5 months later (blank where the model was not in that arm). **Panel B:** one macro indicator moved across the same  $-2 \dots +2$  range with the other four held neutral, so each column is comparable to  $\beta$  above;  $\Sigma/\beta$  sums the five and divides by the joint slope, below one meaning the indicators reinforce one another. <sup>†</sup>interval includes zero.

| <i>Panel A: procyclicality, through-the-cycle response, and revision arms</i> |                    |                    |                    |                       |           |                |               |
|-------------------------------------------------------------------------------|--------------------|--------------------|--------------------|-----------------------|-----------|----------------|---------------|
| Model                                                                         | $\beta$ (95% CI)   | Swing              | D:U                | Resid. (95% CI)       | PD-ret.   | $\beta$ real   | $\Delta\beta$ |
| Gemini Pro                                                                    | 0.182 [0.15, 0.22] | 0.72               | 2.3                | −0.021 [−0.04, 0.00]  | 0.00      |                |               |
| GPT-6 Astra                                                                   | 0.198 [0.18, 0.22] | 0.87               | 4.8                | −0.055 [−0.08, −0.04] | −0.11     | 0.151          |               |
| Gemini Flash                                                                  | 0.230 [0.19, 0.27] | 0.96               | 4.0                | 0.059 [0.03, 0.08]    | 0.22      | 0.094          | −0.037        |
| Gemini Flash-Lite                                                             | 0.285 [0.25, 0.32] | 1.15               | 3.6                | 0.056 [0.03, 0.08]    | 0.34      |                |               |
| GPT-4o                                                                        | 0.387 [0.34, 0.43] | 1.55               | 3.3                | 0.187 [0.15, 0.22]    | 0.52      | 0.407          | +0.002        |
| GPT-4.1-mini                                                                  | 0.471 [0.41, 0.53] | 1.94               | 3.9                | 0.096 [0.06, 0.13]    | 0.24      |                | −0.021        |
| GPT-4o-mini                                                                   | 0.775 [0.69, 0.86] | 3.18               | 2.6                | 0.403 [0.34, 0.47]    | 0.59      | 0.814          | −0.028        |
| <i>Panel B: macro-factor decomposition (one indicator at a time)</i>          |                    |                    |                    |                       |           |                |               |
| Model                                                                         | GDP                | Unem.              | Spread             | Default               | Standards | $\Sigma/\beta$ |               |
| GPT-4o-mini                                                                   | 0.184              | 0.197              | 0.148              | 0.199                 | 0.175     | 1.17           |               |
| GPT-4o                                                                        | 0.019              | 0.043              | 0.112              | 0.114                 | 0.047     | 0.87           |               |
| Gemini Flash                                                                  | 0.029              | 0.017 <sup>†</sup> | 0.015 <sup>†</sup> | 0.046                 | 0.037     | 0.62           |               |
| GPT-6 Astra                                                                   | 0.016              | 0.012 <sup>†</sup> | 0.091              | 0.034                 | 0.035     | 0.95           |               |

where capability is highest: the frontier models produce the most stable-looking rating while carrying the least risk signal. Translated into capital requirements, the direction of the swing survives every mapping, loss-given-default and maturity assumption we vary, though its magnitude does not.

What the experiment demonstrates is a behavioural mechanism in the served endpoints at the collection epoch, not a property of a model family in perpetuity and not an evaluation of deployed credit systems: we observe no rating committee, no lending decision, and no analyst acting on these outputs. Varying the macroeconomic context around fixed firm information does capture an exposure that arises in real pipelines, where models reach macroeconomic information through market data, news, retrieval or analyst inputs. Replacing the synthetic fundamentals with real de-identified filing-derived fundamentals leaves the effect in place, which narrows the realism gap on the firm side but not on the others, and the frontier model's ability to recognise roughly a sixth of those tested, despite name, ticker and date removal, is itself why the primary design is synthetic. Future work could benchmark against agency and internal-rating baselines, and test whether fine-tuning or retrieval design can reduce procyclicality without suppressing genuine changes in credit risk.

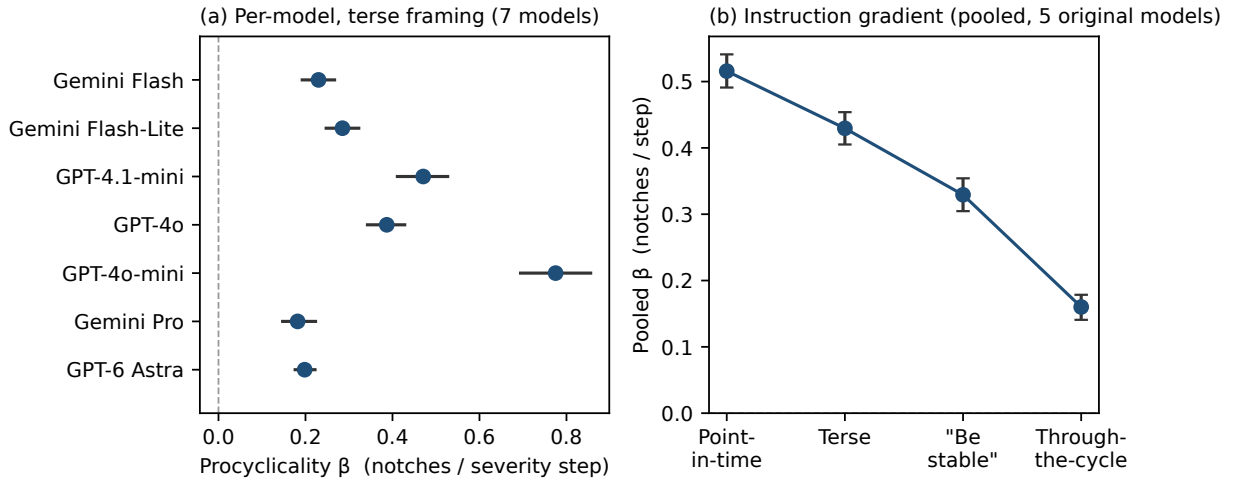
## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist in implementing the data-capture and analysis pipeline and in drafting and editing the manuscript text. The large language models evaluated in this study, four OpenAI models and three Google Gemini models, constitute the object of study of this paper, and their use is fully described in Section 3. They are not tools of manuscript preparation. All AI-assisted content was carefully reviewed, edited, and verified by the authors, who take full responsibility for the content of the published article.

**Table 2**

Pooled procyclicality by prompt framing, notches per severity step with firm-clustered bootstrap 95% CIs. Pooled across the five non-frontier models, whose four framings form the original confirmatory grid; the two frontier models' framing gradients are reported in the replication archive because their arms were added in revision.

| Framing           | Pooled $\beta$ | 95% CI       | Excl. 0 |
|-------------------|----------------|--------------|---------|
| Point-in-time     | 0.516          | [0.49, 0.54] | Yes     |
| Terse             | 0.429          | [0.41, 0.45] | Yes     |
| "Be stable"       | 0.329          | [0.30, 0.35] | Yes     |
| Through-the-cycle | 0.160          | [0.14, 0.18] | Yes     |



**Figure 1:** (a) Per-model procyclicality  $\beta$  under the terse framing, all seven models, with 95% firm-clustered bootstrap CIs. The dashed line marks the through-the-cycle null ( $\beta = 0$ ). (b) Pooled  $\beta$  across the four prompt framings (point-in-time  $\rightarrow$  terse  $\rightarrow$  "be stable"  $\rightarrow$  through-the-cycle) with 95% CIs, pooled across the five original non-frontier models as in Table 2. The two frontier arms enter panel (a) and the per-model tables.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Funding

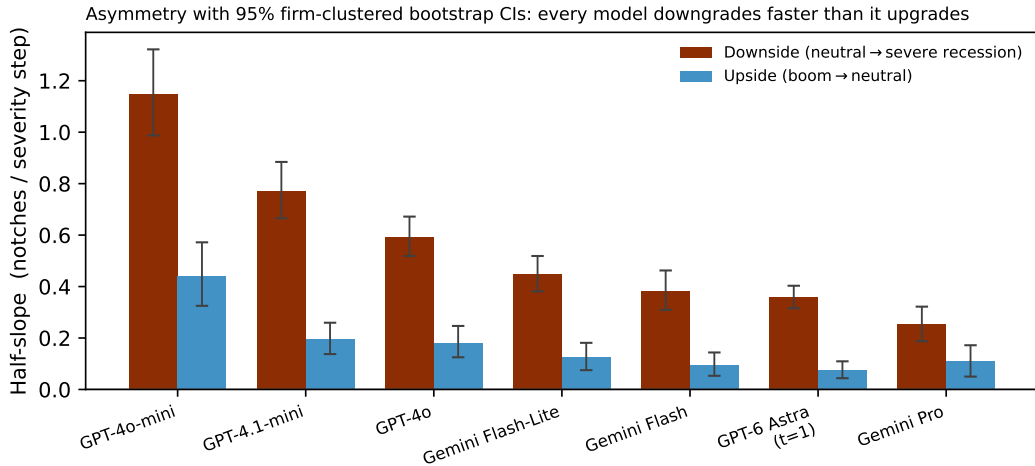
This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

## Data availability

All data and code required to reproduce every result are openly available in a public repository. The citable archive (DOI) is withheld here to preserve author anonymity for double-blind review and will be provided on acceptance (Anonymous Authors, 2026).

## A. Robustness

The original confirmatory grid contains 32,000 assessments. Including the revision experiments, the study contains 80,590 frozen assessments, every one of which parsed. The rating scale has 21 points because the default grades D, SD and RD share the bottom notch with C. D accounts for 90 assessments and SD and RD



**Figure 2:** Asymmetry: per-model downside half-slope (neutral→severe recession) versus upside half-slope (boom→neutral), terse framing, with 95% firm-clustered bootstrap confidence intervals. Every model downgrades faster into recessions than it upgrades into booms, and every downside-to-upside ratio interval excludes 1. The ratio intervals are wide because the upside half-slope is small, so the ordering of models by asymmetry is not resolvable. (t=1) marks the one arm that could not run at temperature zero.

for none. A further 101 call attempts, 0.13% of the 80,691 made in frozen captures, returned a transport or schema error at capture and are excluded from the frozen set. Table 3 reports the synthetic-firm generation specification.

*Placebo and alternative estimators.* The pooled macro slope is 0.4295 against 0.0425 for the matched credit-irrelevant placebo, yielding a placebo-adjusted effect of 0.387 notches per severity step (95% CI [0.363, 0.412]). The interval is obtained by bootstrapping the slope difference on the same resampled firms. A joint specification containing both axes returns the same coefficient and the same interval. Served-model fixed effects shift OpenAI coefficients by less than 0.003. Table 4 reports the expected-notch, replication-noise and ordered-probit estimates.

*Real fundamentals and issuer recall.* Using FY2024 financial data from 80 SEC filers, with names, tickers, CIKs and dates removed before capture, the positive slope replicates in all four models tested: GPT-4o-mini 0.814 [0.729, 0.896], GPT-4o 0.407 [0.362, 0.451], GPT-6 Astra 0.151 [0.126, 0.179] and Gemini Flash 0.094 [0.069, 0.121]. All intervals exclude zero. Of the 45 line-by-tier comparisons between the Table 3 ranges and the real-firm interquartile ranges, 41 overlap. The four exceptions are top-tier revenue and net debt to EBITDA, and distressed-firm current ratio and working capital to assets. Because synthetic lines are drawn independently, this supports marginal credit-ratio realism, not firm scale or the joint covariance structure of real filings. GPT-6 Astra identifies 7/40 de-identified real firms and GPT-4o 0/40, versus 0/40 synthetic controls. The slope difference between identified and tested-but-unidentified firms is  $-0.003$  [ $-0.074, 0.063$ ], with the identified subsample too small to establish equivalence.

*Replication and model drift.* The primary condition has five replicates per cell for `gpt-4o-mini`, `gpt-4.1-mini`, `gpt-4o` and `gemini-flash-latest`, and two for the remaining models. Three OpenAI arms use dated snapshots (`gpt-4o-mini-2024-07-18`, `gpt-4.1-mini-2025-04-14`, `gpt-4o-2024-11-20`) and log `system_fingerprint`, making the served backend identifiable per row but not reproducible later. GPT-6 Astra and the three Gemini `-latest` endpoints use unresolved aliases. Cross-epoch coefficient drift remains within sampling uncertainty.

*Formal contrasts and sensitivity.* Paired firm-clustered bootstraps show that 18 of 21 model pairs differ significantly (16 after a Holm correction). The “be stable” versus through-the-cycle contrast is significant in all seven models, ranging from 0.047 [0.017, 0.080] to 0.254 [0.206, 0.304]. PD-retention intervals lie below the 0.75 threshold for every model, ranging from 0.587 [0.529, 0.642] to  $-0.106$  [ $-0.166$ ,  $-0.059$ ]. Under the “be stable” instruction PD retention instead ranges from 0.42 to 0.92.

The baseline capital mapping uses a stylised 21-notch PD ladder, anchored at approximately AAA 0.01%, BBB 0.25%, BB 1.1%, B 5% and CCC 20%, with LGD = 0.45 and maturity = 2.5 years. The ladder reproduces the ordering and broad magnitudes of the one-year corporate default rates reported by S&P Global Ratings (2025) rather than copying that study’s values. The Basel 0.05% PD input floor of Basel Committee on Banking Supervision (2020b, CRE32.4) is then applied, which binds on the four notches at AA– and above. Reported capital figures are on the floored ladder. A 259-cell sensitivity grid varies four PD mappings (baseline, PD $\times$ 0.5, PD $\times$ 2 and a one-notch downward shift), three LGDs (0.30, 0.45, 0.60), three maturities (1.0, 2.5, 5.0 years) and seven models, plus one verbalised-PD mapping per model. The capital swing remains positive in all 259 cells. Its magnitude varies by 2.7 to 4.7 times across the rating-based specifications and by 3.1 to 7.6 times including verbalised PD. The  $\pm 0.05$ -notch TTC rating band and 0.75 PD-retention threshold are reporting conventions rather than supervisory standards.

**Table 3**

Synthetic-firm generation specification. Each firm draws every line independently and uniformly from the interval shown for its credit-quality tier; firms are stratified across the five tiers and eight sectors so the fundamentals-implied ratings span the scale. Ranges were chosen so a tier's implied rating lands in its band, not fitted to a reference population. Of the 45 line-by-tier cells, 41 have a draw interval intersecting the interquartile range of the real firms in that tier (Appendix A). Values are frozen and SHA-256'd before capture.

| Line                       | High IG         | Low IG        | Crossover     | High yield    | Distressed     |
|----------------------------|-----------------|---------------|---------------|---------------|----------------|
| Revenue (USD m)            | 40000 to 180000 | 8000 to 60000 | 2000 to 20000 | 500 to 8000   | 100 to 3000    |
| Revenue growth (%)         | 2 to 9          | 1 to 7        | -2 to 6       | -5 to 6       | -18 to 2       |
| EBITDA margin (%)          | 22 to 38        | 15 to 27      | 11 to 20      | 7 to 16       | 0 to 9         |
| Net debt / EBITDA (×)      | 0.2 to 1.3      | 1.3 to 2.8    | 2.8 to 4.2    | 4.2 to 6.5    | 6.5 to 12.0    |
| EBITDA / interest (×)      | 14.0 to 40.0    | 6.0 to 13.0   | 3.0 to 6.0    | 1.6 to 3.2    | 0.4 to 1.6     |
| FCF margin (%)             | 9 to 18         | 4 to 10       | 0 to 6        | -4 to 3       | -14 to -1      |
| Current ratio              | 1.4 to 2.4      | 1.1 to 1.8    | 0.9 to 1.4    | 0.7 to 1.2    | 0.4 to 0.9     |
| Retained earnings / assets | 0.35 to 0.60    | 0.18 to 0.40  | 0.05 to 0.22  | -0.05 to 0.10 | -0.40 to -0.02 |
| Working capital / assets   | 0.15 to 0.32    | 0.06 to 0.20  | -0.02 to 0.12 | -0.08 to 0.06 | -0.25 to -0.02 |

**Table 4**

Robustness of the terse-framing coefficient. Net = placebo-adjusted coefficient. Exp-n. = expected-notch slope from rating-token log-probabilities. Noise SD = mean within-cell notch standard deviation across replicate seeds. Probit = Amato-Furfine ordered-probit estimate, converted from the latent scale to notches per step. Expected-notch estimates are unavailable for Gemini and for GPT-6 Astra because neither returns rating-token log-probabilities. Gemini Flash-Lite's 0.000 Noise SD is exact.

| Model             | $\beta$ | Net   | Exp-n. terse | Exp-n. TTC | Noise SD | Probit |
|-------------------|---------|-------|--------------|------------|----------|--------|
| Gemini Pro        | 0.182   | 0.174 |              |            | 0.080    | 0.138  |
| GPT-6 Astra       | 0.198   | 0.186 |              |            | 0.067    | 0.177  |
| Gemini Flash      | 0.230   | 0.190 |              |            | 0.020    | 0.272  |
| Gemini Flash-Lite | 0.285   | 0.268 |              |            | 0.000    | 0.235  |
| GPT-4o            | 0.387   | 0.367 | 0.378        | 0.182      | 0.025    | 0.403  |
| GPT-4.1-mini      | 0.471   | 0.464 | 0.420        | 0.081      | 0.086    | 0.307  |
| GPT-4o-mini       | 0.775   | 0.646 | 0.765        | 0.418      | 0.072    | 0.726  |

## B. Prompts

The prompt framings and the macro and placebo blocks below are extracted programmatically from the capture code (`agent.py`) and reproduce the executed instrument verbatim.

### B.1. Prompt framings

The four instruction framings (system prompt), verbatim as executed:

You are a credit rating analyst. Assign a long-term issuer credit rating to the company described, based on its fundamentals and the stated operating environment. Respond with ONLY a JSON object and nothing else, in exactly this form: {"rating": "<one S&P long-term issuer rating from AAA down to D, using +/- modifiers, e.g. AA-, BBB+, B>", "pd\_1yr\_bps": <your estimated 1-year probability of default in BASIS POINTS as an integer, where 100 bps = 1%>, "rationale": "<=30 words>"}

You are a credit rating analyst applying a THROUGH-THE-CYCLE methodology. Assign a long-term issuer credit rating that reflects the company's creditworthiness over a full economic cycle, deliberately LOOKING THROUGH transitory macroeconomic conditions rather than the current point in the cycle. Respond with ONLY a JSON object and nothing else, in exactly this form: {"rating": "<one S&P long-term issuer rating from AAA down to D, using +/- modifiers, e.g. AA-, BBB+, B>", "pd\_1yr\_bps": <your estimated 1-year probability of default in BASIS POINTS as an integer, where 100 bps = 1%>, "rationale": "<=30 words>"}

## Does AI rate through the cycle?

You are a credit rating analyst applying a POINT-IN-TIME methodology. Assign a long-term issuer credit rating that reflects the company's creditworthiness given CURRENT economic and financial conditions as described. Respond with ONLY a JSON object and nothing else, in exactly this form: {"rating": "<one S&P long-term issuer rating from AAA down to D, using +/- modifiers, e.g. AA-, BBB+, B>", "pd\_1yr\_bps": <your estimated 1-year probability of default in BASIS POINTS as an integer, where 100 bps = 1%>, "rationale": "<=30 words">}.</p>
</div>
<div data-bbox="67 177 925 222" data-label="Text">
<p>You are a credit rating analyst. Assign a long-term issuer credit rating that is STABLE and does not change much in response to short-term conditions. Respond with ONLY a JSON object and nothing else, in exactly this form: {"rating": "<one S&P long-term issuer rating from AAA down to D, using +/- modifiers, e.g. AA-, BBB+, B>", "pd\_1yr\_bps": <your estimated 1-year probability of default in BASIS POINTS as an integer, where 100 bps = 1%>, "rationale": "<=30 words">}.</p>
</div>
<div data-bbox="67 260 386 276" data-label="Section-Header">
<h2>B.2. Macro-environment blocks</h2>
</div>
<div data-bbox="94 276 895 292" data-label="Text">
<p>Prepended to the byte-identical fundamentals; only these five blocks differ across the severity axis:</p>
</div>
<div data-bbox="67 327 472 340" data-label="Text">
<p>MACRO ENVIRONMENT (current conditions, this fiscal period):</p>
</div>
<div data-bbox="80 338 424 394" data-label="List-Group">
<ul>
<li>- Real GDP growth (annualized): +4.5%</li>
<li>- Unemployment rate: 3.4%</li>
<li>- Investment-grade corporate credit spread: 95 bps</li>
<li>- Trailing 12-month corporate default rate: 0.6%</li>
<li>- Bank lending standards: easing broadly</li>
</ul>
</div>
<div data-bbox="67 437 472 450" data-label="Text">
<p>MACRO ENVIRONMENT (current conditions, this fiscal period):</p>
</div>
<div data-bbox="80 448 430 504" data-label="List-Group">
<ul>
<li>- Real GDP growth (annualized): +2.6%</li>
<li>- Unemployment rate: 4.2%</li>
<li>- Investment-grade corporate credit spread: 135 bps</li>
<li>- Trailing 12-month corporate default rate: 1.4%</li>
<li>- Bank lending standards: modestly easing</li>
</ul>
</div>
<div data-bbox="67 548 472 560" data-label="Text">
<p>MACRO ENVIRONMENT (current conditions, this fiscal period):</p>
</div>
<div data-bbox="80 559 430 613" data-label="List-Group">
<ul>
<li>- Real GDP growth (annualized): +1.8%</li>
<li>- Unemployment rate: 5.0%</li>
<li>- Investment-grade corporate credit spread: 175 bps</li>
<li>- Trailing 12-month corporate default rate: 2.3%</li>
<li>- Bank lending standards: unchanged</li>
</ul>
</div>
<div data-bbox="67 657 472 669" data-label="Text">
<p>MACRO ENVIRONMENT (current conditions, this fiscal period):</p>
</div>
<div data-bbox="80 668 430 723" data-label="List-Group">
<ul>
<li>- Real GDP growth (annualized): -0.8%</li>
<li>- Unemployment rate: 6.4%</li>
<li>- Investment-grade corporate credit spread: 320 bps</li>
<li>- Trailing 12-month corporate default rate: 4.1%</li>
<li>- Bank lending standards: tightening</li>
</ul>
</div>
<div data-bbox="67 767 472 779" data-label="Text">
<p>MACRO ENVIRONMENT (current conditions, this fiscal period):</p>
</div>
<div data-bbox="80 777 430 832" data-label="List-Group">
<ul>
<li>- Real GDP growth (annualized): -3.6%</li>
<li>- Unemployment rate: 8.9%</li>
<li>- Investment-grade corporate credit spread: 620 bps</li>
<li>- Trailing 12-month corporate default rate: 7.5%</li>
<li>- Bank lending standards: sharply tightening</li>
</ul>
</div>
<div data-bbox="67 869 528 887" data-label="Section-Header">
<h2>B.3. Placebo blocks (credit-irrelevant control)</h2>
</div>
<div data-bbox="94 887 653 903" data-label="Text">
<p>Matched in structure and number density, on a credit-irrelevant axis:</p>
</div>
<div data-bbox="67 943 307 959" data-label="Page-Footer">
<p>: Preprint submitted to Elsevier</p>
</div>
<div data-bbox="822 943 939 959" data-label="Page-Footer">
<p>Page 10 of 12</p>
</div>

## Does AI rate through the cycle?

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: +0.6 degC
- Daylight hours vs annual mean: +4.5%
- Monthly rainfall vs normal: 95 mm
- Regional air-quality index: 38
- Commuter traffic congestion: light

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: +0.2 degC
- Daylight hours vs annual mean: +2.6%
- Monthly rainfall vs normal: 135 mm
- Regional air-quality index: 52
- Commuter traffic congestion: moderate

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: 0.0 degC
- Daylight hours vs annual mean: +1.8%
- Monthly rainfall vs normal: 175 mm
- Regional air-quality index: 60
- Commuter traffic congestion: typical

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: -0.4 degC
- Daylight hours vs annual mean: -0.8%
- Monthly rainfall vs normal: 320 mm
- Regional air-quality index: 88
- Commuter traffic congestion: heavy

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: -1.1 degC
- Daylight hours vs annual mean: -3.6%
- Monthly rainfall vs normal: 620 mm
- Regional air-quality index: 140
- Commuter traffic congestion: severe

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